Table B3. Spline-indicator fixed-parameter binary logit model of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-vehicle crashes without a motorcycle, Thailand, 2016–2023: implied parameter for each year, estimated parameter (*t*-statistic).

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Constant [serious injury / fatal]									
Constant	1, 2, 3, 8	–1.524 (–17.06)	–1.900 (–22.25)	–2.093 (–29.18)	–2.287 (–33.99)	–2.480 (–33.66)	–2.673 (–30.16)	–2.867 (–26.39)	–2.623 (–19.41)
Vehicle involvement									
One passenger car involved (1 if yes, 0 otherwise)	1, 6	0.039 (4.28)	0.078 (4.28)	0.117 (4.28)	0.156 (4.28)	0.195 (4.28)	0.165 (4.78)	0.134 (3.09)	0.104 (1.60)
One 6-wheel truck involved (1 if yes, 0 otherwise)	1, 2	–0.392 (–3.46)	–0.370 (–4.02)	–0.347 (–4.67)	–0.325 (–5.08)	–0.303 (–4.73)	–0.281 (–3.76)	–0.258 (–2.80)	–0.236 (–2.08)
One 10-wheel truck involved (1 if yes, 0 otherwise)	1, 2	–0.729 (–5.43)	–0.686 (–6.30)	–0.643 (–7.27)	–0.600 (–7.84)	–0.558 (–7.20)	–0.515 (–5.68)	–0.472 (–4.22)	–0.429 (–3.12)
One truck with more than 10 wheels involved (1 if yes, 0 otherwise)	1, 2	–0.858 (–8.99)	–0.809 (–10.43)	–0.760 (–12.13)	–0.711 (–13.31)	–0.662 (–12.51)	–0.613 (–9.99)	–0.564 (–7.44)	–0.515 (–5.50)
Van involved (1 if yes, 0 otherwise)	1, 2	0.693 (5.43)	0.671 (6.56)	0.648 (7.76)	0.626 (8.13)	0.604 (7.08)	0.581 (5.54)	0.559 (4.27)	0.537 (3.35)
Pedestrian involved (1 if yes, 0 otherwise)	1, 2, 8	3.687 (18.46)	3.671 (23.15)	3.654 (29.26)	3.638 (34.35)	3.621 (33.02)	3.605 (26.85)	3.588 (21.00)	4.647 (15.57)
Crash type									
Rear-end collision in the same lane (1 if yes, 0 otherwise)	1, 2	0.744 (4.46)	0.793 (5.84)	0.842 (7.40)	0.891 (8.31)	0.940 (7.94)	0.988 (6.90)	1.037 (5.89)	1.086 (5.09)
Ran off road to the left and hit an object (1 if yes, 0 otherwise)	1, 2, 6	–0.608 (–6.03)	–0.464 (–6.05)	–0.319 (–5.16)	–0.175 (–2.75)	–0.031 (–0.38)	–0.099 (–1.39)	–0.167 (–1.97)	–0.235 (–2.07)
Vehicle mounted a traffic island (1 if yes, 0 otherwise)	1, 2, 8	–0.665 (–6.71)	–0.620 (–7.86)	–0.576 (–8.80)	–0.532 (–8.46)	–0.487 (–6.74)	–0.443 (–4.91)	–0.399 (–3.55)	–0.803 (–5.48)
Ran off road to the right and hit an object (1 if yes, 0 otherwise)	1, 2, 8	–0.476 (–4.86)	–0.409 (–5.21)	–0.342 (–5.34)	–0.275 (–4.70)	–0.208 (–3.24)	–0.141 (–1.79)	–0.074 (–0.75)	–0.526 (–3.79)
Ran off road to the left (1 if yes, 0 otherwise)	1, 2, 7, 8	0.471 (4.84)	0.488 (6.40)	0.505 (7.98)	0.522 (8.17)	0.539 (6.93)	0.556 (5.59)	0.928 (7.03)	0.677 (4.64)
Ran off road on a right bend and hit an object (1 if yes, 0 otherwise)	1, 4, 5, 6, 7, 8	–0.300 (–4.39)	–0.600 (–4.39)	–0.901 (–4.39)	0.059 (0.27)	0.176 (1.02)	–0.482 (–2.64)	–0.021 (–0.12)	–0.352 (–1.70)
Ran off road on a left bend and hit an object (1 if yes, 0 otherwise)	1, 4, 5	–0.287 (–4.14)	–0.573 (–4.14)	–0.860 (–4.14)	–0.135 (–0.63)	–0.136 (–0.88)	–0.138 (–1.13)	–0.140 (–1.02)	–0.142 (–0.76)
Ran off road to the right (1 if yes, 0 otherwise)	1, 2	0.608 (5.94)	0.586 (7.06)	0.565 (8.18)	0.543 (8.50)	0.522 (7.49)	0.501 (5.95)	0.479 (4.63)	0.458 (3.64)
Ran off road on a left bend (1 if yes, 0 otherwise)	1, 2, 4, 5, 7	0.537 (3.17)	–0.048 (–0.32)	–0.634 (–2.64)	0.870 (3.61)	0.450 (2.67)	0.031 (0.15)	0.169 (1.03)	0.306 (1.22)
Ran off road on a right bend (1 if yes, 0 otherwise)	1, 2, 4, 5	0.489 (2.80)	–0.039 (–0.26)	–0.568 (–2.35)	0.426 (1.84)	0.443 (2.66)	0.461 (3.39)	0.479 (3.00)	0.497 (2.25)
Crash cause									
Speeding (1 if yes, 0 otherwise)	1	0.081 (8.56)	0.162 (8.56)	0.243 (8.56)	0.324 (8.56)	0.405 (8.56)	0.486 (8.56)	0.567 (8.56)	0.648 (8.56)
Cutting in too closely (1 if yes, 0 otherwise)	1, 4	–0.014 (–0.25)	–0.027 (–0.25)	–0.041 (–0.25)	0.142 (1.11)	0.325 (2.86)	0.508 (3.92)	0.692 (4.14)	0.875 (4.06)
Driver fell asleep (1 if yes, 0 otherwise)	1, 2, 6, 7	0.512 (6.49)	0.583 (10.01)	0.655 (12.86)	0.726 (11.74)	0.798 (9.48)	0.421 (3.49)	0.668 (7.55)	0.915 (8.12)
Roadway characteristics									
Crash on the main carriageway (1 if yes, 0 otherwise)	1, 2	–0.590 (–5.24)	–0.516 (–5.66)	–0.441 (–5.95)	–0.367 (–5.64)	–0.293 (–4.36)	–0.219 (–2.75)	–0.145 (–1.47)	–0.071 (–0.59)
Road under construction (1 if yes, 0 otherwise)	1, 3, 5	–0.101 (–1.02)	–0.203 (–1.02)	0.226 (1.81)	0.654 (3.72)	0.483 (3.88)	0.312 (2.96)	0.142 (1.05)	–0.029 (–0.15)
Painted median (1 if yes, 0 otherwise)	1, 3, 5, 6	–0.210 (–3.02)	–0.420 (–3.02)	–0.171 (–2.02)	0.079 (0.66)	–0.363 (–2.80)	–0.266 (–2.97)	–0.169 (–1.73)	–0.071 (–0.49)
Raised or earth median (1 if yes, 0 otherwise)	1, 2	–0.217 (–3.17)	–0.206 (–3.71)	–0.195 (–4.30)	–0.184 (–4.58)	–0.173 (–4.13)	–0.162 (–3.25)	–0.151 (–2.45)	–0.140 (–1.85)
Depressed median (1 if yes, 0 otherwise)	1, 2, 4	–0.134 (–1.85)	–0.053 (–1.12)	0.029 (0.52)	0.002 (0.05)	–0.024 (–0.63)	–0.051 (–1.17)	–0.077 (–1.39)	–0.103 (–1.45)
Barrier median (1 if yes, 0 otherwise)	1, 2	–0.598 (–6.17)	–0.655 (–8.33)	–0.712 (–11.25)	–0.770 (–14.38)	–0.827 (–15.77)	–0.884 (–14.61)	–0.941 (–12.57)	–0.998 (–10.78)
Horizontal curve (1 if yes, 0 otherwise)	1, 2, 4, 5, 7	–0.260 (–2.56)	0.135 (1.15)	0.530 (2.83)	–0.314 (–1.59)	–0.064 (–0.46)	0.186 (1.37)	0.031 (0.26)	–0.125 (–0.76)
Concrete pavement surface (1 if yes, 0 otherwise)	1	–0.006 (–0.70)	–0.012 (–0.70)	–0.017 (–0.70)	–0.023 (–0.70)	–0.029 (–0.70)	–0.035 (–0.70)	–0.041 (–0.70)	–0.046 (–0.70)
Crest vertical curve (1 if yes, 0 otherwise)	1	0.009 (0.48)	0.017 (0.48)	0.026 (0.48)	0.035 (0.48)	0.043 (0.48)	0.052 (0.48)	0.061 (0.48)	0.070 (0.48)
Sag vertical curve (1 if yes, 0 otherwise)	1, 4, 6	0.104 (1.60)	0.208 (1.60)	0.312 (1.60)	–0.033 (–0.24)	–0.378 (–1.68)	–0.141 (–0.94)	0.096 (0.58)	0.333 (1.32)
On grade (1 if yes, 0 otherwise)	1	–0.003 (–0.23)	–0.006 (–0.23)	–0.009 (–0.23)	–0.012 (–0.23)	–0.015 (–0.23)	–0.018 (–0.23)	–0.021 (–0.23)	–0.024 (–0.23)
T-intersection (1 if yes, 0 otherwise)	1, 6, 8	–0.126 (–3.36)	–0.253 (–3.36)	–0.379 (–3.36)	–0.506 (–3.36)	–0.632 (–3.36)	–0.340 (–2.80)	–0.047 (–0.26)	–0.457 (–2.13)
U-turn with deceleration lane only (1 if yes, 0 otherwise)	1, 4, 5, 6	–0.146 (–1.69)	–0.293 (–1.69)	–0.439 (–1.69)	0.594 (2.45)	–0.280 (–1.30)	–0.239 (–1.71)	–0.199 (–1.52)	–0.158 (–0.81)
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise)	1, 8	–0.056 (–2.16)	–0.112 (–2.16)	–0.168 (–2.16)	–0.224 (–2.16)	–0.280 (–2.16)	–0.336 (–2.16)	–0.392 (–2.16)	0.249 (0.94)
Bridge section (1 if yes, 0 otherwise)	1, 2, 3	–0.161 (–0.59)	0.611 (3.90)	0.581 (4.77)	0.551 (5.63)	0.522 (5.55)	0.492 (4.39)	0.462 (3.21)	0.433 (2.37)
Environmental characteristics									
Wet road surface (1 if yes, 0 otherwise)	1, 5	–0.153 (–2.41)	–0.306 (–2.41)	–0.459 (–2.41)	–0.612 (–2.41)	–0.491 (–2.71)	–0.370 (–2.70)	–0.249 (–1.65)	–0.128 (–0.61)
Raining (1 if yes, 0 otherwise)	1, 3, 5, 6	–0.050 (–0.68)	–0.100 (–0.68)	0.130 (0.67)	0.361 (1.36)	–0.022 (–0.11)	–0.069 (–0.48)	–0.116 (–0.73)	–0.163 (–0.73)
Dark, no street lighting (1 if yes, 0 otherwise)	1, 2	0.450 (5.35)	0.417 (5.93)	0.384 (6.15)	0.351 (5.59)	0.318 (4.46)	0.285 (3.33)	0.252 (2.44)	0.219 (1.79)

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Dark, with street lighting (1 if yes, 0 otherwise)	1, 5, 7	–0.041 (–2.10)	–0.082 (–2.10)	–0.123 (–2.10)	–0.164 (–2.10)	–0.057 (–0.80)	0.051 (0.56)	0.031 (0.33)	0.011 (0.10)
Temporal characteristics									
Weekend (1 if yes, 0 otherwise)	1	0.011 (2.13)	0.023 (2.13)	0.034 (2.13)	0.045 (2.13)	0.057 (2.13)	0.068 (2.13)	0.079 (2.13)	0.090 (2.13)
Songkran festival period (1 if yes, 0 otherwise)	1	0.009 (0.82)	0.019 (0.82)	0.028 (0.82)	0.037 (0.82)	0.046 (0.82)	0.056 (0.82)	0.065 (0.82)	0.074 (0.82)
New Year festival period (1 if yes, 0 otherwise)	1, 8	0.030 (2.40)	0.059 (2.40)	0.089 (2.40)	0.119 (2.40)	0.149 (2.40)	0.178 (2.40)	0.208 (2.40)	–0.175 (–1.04)
Public holiday (1 if yes, 0 otherwise)	1	–0.028 (–1.61)	–0.057 (–1.61)	–0.085 (–1.61)	–0.113 (–1.61)	–0.142 (–1.61)	–0.170 (–1.61)	–0.198 (–1.61)	–0.227 (–1.61)
Evening (1 if yes, 0 otherwise)	1, 3	0.109 (2.52)	0.219 (2.52)	0.202 (2.76)	0.185 (2.74)	0.168 (2.35)	0.151 (1.80)	0.135 (1.31)	0.118 (0.95)
Midnight / early morning (1 if yes, 0 otherwise)	1, 3	0.163 (4.24)	0.326 (4.24)	0.285 (4.42)	0.244 (4.09)	0.204 (3.18)	0.163 (2.14)	0.123 (1.32)	0.082 (0.73)
Peak hours (1 if yes, 0 otherwise)	1, 3, 5, 7	0.069 (1.66)	0.138 (1.66)	0.026 (0.46)	–0.087 (–1.09)	0.020 (0.38)	0.127 (1.58)	0.024 (0.40)	–0.079 (–0.84)
Region									
Eastern region (1 if yes, 0 otherwise)	1	0.027 (3.08)	0.054 (3.08)	0.081 (3.08)	0.109 (3.08)	0.136 (3.08)	0.163 (3.08)	0.190 (3.08)	0.217 (3.08)
Northeastern region (1 if yes, 0 otherwise)	1, 4, 6	–0.016 (–0.72)	–0.033 (–0.72)	–0.049 (–0.72)	0.083 (1.76)	0.215 (3.26)	0.139 (2.80)	0.064 (1.13)	–0.012 (–0.15)
Northern region (1 if yes, 0 otherwise)	1, 2, 4	–0.192 (–1.88)	–0.058 (–0.93)	0.077 (1.01)	0.011 (0.19)	–0.055 (–1.13)	–0.122 (–2.25)	–0.188 (–2.69)	–0.254 (–2.79)
Southern region (1 if yes, 0 otherwise)	1, 3, 5	0.133 (3.39)	0.265 (3.39)	0.152 (2.91)	0.038 (0.53)	0.096 (1.79)	0.154 (3.12)	0.212 (3.46)	0.270 (3.27)
Western region (1 if yes, 0 otherwise)	1, 2, 6	–0.036 (–0.40)	0.152 (2.25)	0.340 (6.42)	0.528 (9.15)	0.716 (9.17)	0.533 (8.95)	0.349 (4.73)	0.166 (1.52)
Model statistics									
Number of observations					73,731				
Number of estimated spline coefficients					135				
Variables with the base term only (parameter proportional to the period index)					8 of 51				
Log-likelihood at zero, LL(0)					–51106.43				
Log-likelihood with constant only, LL(C)					–26810.27				
Log-likelihood at convergence, LL(β)					–23653.82				
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$					0.537				
AIC / BIC					47,577.6 / 48,820.7				
LR test against the fixed-parameter separate-period models					$\chi^2 = 181.18$, df = 273, confidence of rejection = 0.000				

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one pickup truck. Following Shabab et al. (2024), each variable enters through spline terms $f_j(\tau) = \max(\tau - j + 1, 0)$, $j = 1, \ldots, 8$, where $\tau = 1, \ldots, 8$ indexes the years; the parameter in year τ is $\beta(\tau) = \Sigma_j b_j f_j(\tau)$. The base term ($j = 1$) is always retained; the other terms were removed by backward elimination at the 90% level ($|t| < 1.645$, one term per variable per round). A variable with the base term only has a parameter proportional to the period index. The table reports the implied parameters $\beta(\tau)$; *t*-statistics by the delta method from the covariance of the spline coefficients.